

ALFIYA ANSARI

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SUMMARY

Data Analyst with internship and research experience in predictive analytics, machine learning, and data visualization. Experienced in analyzing **1M+ records**, developing forecasting models, and building interactive dashboards using **Python, SQL and Tableau** to support data-driven decision-making.

EXPERIENCE

Research Assistant (AI & Data Science) – University of Koblenz

 Oct 2025 – Present

 Koblenz, Germany

- Designed and implemented an end-to-end ML workflow for heart disease classification, benchmarking **10 ML/DL models** and achieving up to **83% accuracy** and **0.89 ROC-AUC**.
- Engineered clinical and multivariate time-series datasets using **Python, Numpy, Pandas, Seaborn, Scikit-learn, and PyTorch**, enabling predictive modeling across **3 research domains**.
- Developed forecasting pipelines using **30–60 day sliding windows**, recursive forecasting, and data augmentation, generating **30-step ahead predictions**.
- Built predictive models on the **3-dimensional Lorenz System**, evaluating performance under chaotic and non-linear dynamical conditions and produced **7+ analytical visualizations and performance reports**.

Data Science Intern – AB Technologies

 Jan 2023 – Mar 2023

 Nagercoil, India

- Processed and analyzed **1M+ transaction records spanning 7 years (2015–2022)** across a **3-branch restaurant chain**, leveraging SQL, Python, PySpark, Tableau, and machine learning for business analysis and demand forecasting.
- Evaluated demand patterns across **100+ Indian food items** and **3 customer markets**, identifying branch-specific preferences, delivery trends, and profitability drivers.
- Developed a **TensorFlow-based meal-wise forecasting model** achieving **85% prediction accuracy** for food demand, inventory planning, and revenue forecasting.
- Designed and delivered **10+ Tableau dashboards and analytical reports**, and presented findings to stakeholders, providing actionable insights for staffing allocation, menu optimization, and online delivery expansion.

Volunteer Tutor – Rohini College of Engineering and Technology

 Mar 2022 – Dec 2022

 Kanyakumari, India

- Delivered **6 hours of weekly instruction over 10 months** to **50+ students** in Python, Data Analysis, Data Science, and Statistical Analysis, combining theoretical concepts with hands-on practical exercises.

EDUCATION

Masters in Data Science

Universität Koblenz | Grade 2.1

 Oct 2024 – Present

Bachelors in Computer Science

Anna university | Grade 1.7

 Aug 2019 – July 2023

CERTIFICATIONS

- Google Data Analytics Professional Certificate 
- Deep Learning Specialization (DeepLearning.AI) 
- Applied Data Science Program (365 Careers) 
- Machine Learning with Python (Udemy) 
- SQL for Business Intelligence (365 Careers) 

SKILLS

Python SQL R NoSQL

Data Analysis Business Intelligence

Tableau Exploratory Data Analysis

Power BI Excel Pandas

NumPy A/B Testing KPI Analysis

Statistical Analysis MongoDB

Hypothesis Testing PySpark

Data Visualization Data Cleaning

Machine Learning Scikit-learn

Deep Learning TensorFlow

LLMs Dashboard Development

- Designed assignments, coding exercises, and project-based learning activities covering **8+ data science topics**, including data cleaning and preprocessing, visualization, machine learning, deep learning, regression, and classification.

THESIS (ONGOING)

Hybrid Explainable Framework for Automated EHR Data Repair

📅 Expected Oct 2026

📍 Universität Koblenz

- Designed a hybrid framework that automatically transforms the corrupted EHR datasets into reliable datasets using **3 repair approaches**: Rule-Based, Statistical, and LLM-Based methods.
- Developed an LLM-driven routing mechanism to detect and repair **8 health care data quality issues** including missing values, temporal inconsistencies, duplicates, outliers and semantic conflicts.
- Implemented a **4-component architecture** comprising Data Processing, Method Selection, Automated Repair, and Evaluation modules for scalable data repair.
- Evaluated framework performance using **6 metrics**: Repair Accuracy, Ground Truth Comparison, Method Selection Accuracy, Precision, Recall, and F1-Score, while applying Explainable AI principles.

PROJECTS

UAE Real Estate Market Intelligence Dashboard 🔗

- Processed and analyzed **41K+ UAE property listings** using SQL, Python, and Pandas, performing data cleaning, transformation, and feature engineering for large-scale real estate market analysis.
- Developed an interactive **Tableau business intelligence dashboard** analyzing **AED 156.7B in property sales** across multiple cities, enabling KPI tracking and market performance monitoring.
- Performed pricing and market trend analysis, identifying an average property value of **AED 3.79M**, a **2022 price peak of AED 13.6M**, and luxury penthouse dominance with an average value of **AED 23.46M**.
- Conducted comparative city-level analysis and data visualization, generating actionable insights to support real estate market intelligence and investment decision-making.

German Real Estate Fair & Overpricing Detection System 🔗

- Developed a machine learning-based property valuation system for **46K+ German residential property listings** using XGBoost and LightGBM with the Quantile method to estimate fair value and identify overpricing patterns.
- Engineered and optimized **231 predictive features** through data imputation, preprocessing, encoding, and feature transformation to improve accuracy.
- Applied **5-fold cross-validation** and comparative model benchmarking to evaluate multiple machine learning algorithms, ensuring robust and reliable property price estimation.
- Built quantile prediction intervals (**P10–P90**) with **99.78% coverage**, enabling uncertainty-aware pricing analysis, overpricing detection, and risk-informed investment decisions.
- Built and deployed a **Streamlit-based web application** that allows users to estimate fair property values, detect overpricing, and explore uncertainty aware pricing insights for German residential properties.

COMPETENCIES

- Business intelligence reporting and KPI-driven decision making
- Data analysis, validation, and quality assurance
- Statistical analysis and testing
- Visualization, dashboard design, and stakeholder reporting
- Data cleaning, preprocessing, and feature engineering
- Predictive analytics, forecasting, and time-series analysis
- Machine learning, deep learning, and Explainable AI solutions
- Research-driven problem solving and insight generation

LANGUAGES

- English – C1
- German – Learning
- Tamil – Native
- Hindi – Intermediate

ACHIEVEMENTS

- Recognized for outstanding academic performance on **3 occasions as a top-performing student**.
- Elected **Student Council Leader** (05/2022 – 05/2023), leading initiatives and coordinating various events.
- Served as **Captain of Blue House**, leading college sports teams to many successful tournament participation.
- Competed in **inter-college hackathons and technical events**, applying problem solving and teamwork skills.

STRENGTHS

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|---------------------|---------------|
| Analytical Thinking | Teamwork |
| Critical Thinking | Adaptability |
| Research Mindset | Leadership |
| Problem solving | Communication |